3D Thermal Modeling and Optimization of a Radiant-Lamp Curing Process

V2 - Orthotropic heat transfer, cure kinetics, convergence study and residual-stress indicator

Synthetic portfolio project - non-confidential data

July 2026

Abstract

This report presents a synthetic three-dimensional radiant-curing model inspired by aerospace manufacturing process observations. The objective is to move beyond a purely thermal proof of concept by coupling a 3D orthotropic heat-transfer model, infrared/radiant boundary conditions, Kamal-Sourour cure kinetics, exothermic reaction heat, numerical convergence checks, continuous process optimization, and a first-order residual-stress indicator. All parameters are synthetic and non-confidential. The project is positioned as a Mechanical Engineering / thermal-process engineering demonstrator, not as a validated industrial digital twin.

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1 Executive summary

The V1 model already introduced a 3D transient thermal field coupled with a cure model. However, several weaknesses had to be corrected before using the project as a serious Mechanical Engineering portfolio item: no convergence study, ambiguous "anisotropic" terminology, an active zone not sufficiently formalized, a grid-only optimization, and no link to solid-mechanics quantities such as residual strain or stress.

V2 addresses these issues as follows:

- the material is now consistently described as a simplified orthotropic or transversely isotropic thermal medium with $k_x = k_y \neq k_z$;
- the active zone is defined geometrically before simulation as a central bonded/functional ellipse;
- the absorption efficiency η appears explicitly in the parameter table;
- a mesh/time-step convergence study is performed on coarse, medium and fine cases;
- a coarse grid search is used only as initialization, then a continuous SLSQP optimization refines incident flux and lamp-on time;
- a first-order CHILE-like residual-stress indicator is added from thermal strain, cure shrinkage and cure-dependent stiffness.

The selected synthetic cycle on the default medium mesh is approximately:

- incident flux: $q_0 \approx 4095\,\mathrm{W\,m^{-2}}$;
- lamp-on duration: $t_{on} \approx 44.5\,\mathrm{min}$;
- absorbed energy: $\approx 7871\,\mathrm{kJ\,m^{-2}}$;
- maximum cycle temperature: $\approx 115\,^{\circ}\mathrm{C}$;
- maximum centerline through-thickness gradient: $\approx 19.9\,^{\circ}\mathrm{C}$;
- mean final degree of cure: $\bar{\alpha} \approx 0.902$;
- active-zone minimum degree of cure: $\alpha_{active,min} \approx 0.893$;
- global minimum degree of cure: $\alpha_{global,min} \approx 0.631$;
- active-zone residual stress indicator: $\approx 2.7\,\mathrm{MPa}$.

The difference between active-zone and global minimum cure is a key engineering point: the active zone is the pre-defined central bonded patch, whereas the global minimum includes cooler corners outside the functional heating footprint.

2 Project positioning and scope

This project is not a complete industrial cure simulation. It does not reproduce any proprietary geometry, material card, process instruction, tooling, thermocouple map or supplier data. It is a synthetic model whose purpose is to demonstrate the ability to translate a manufacturing observation into a physics-based computational model.

The project is positioned in the following domain:

Dimension	Role in the project
Heat transfer	Transient 3D conduction, convection losses and thermal radiation losses.
Materials	Thermoset-like cure kinetics and exothermic reaction source.
Mechanical Engineering	Process-induced temperature gradients, cure heterogeneity and first-order residual stress risk.
Optimization	Continuous selection of incident flux and lamp-on duration under process constraints.
Validation thinking	Thermocouples, infrared camera, DSC/DMA and mechanical testing as future validation routes.

A more precise statement is therefore:

> This project demonstrates a thermal-process engineering approach applied to a composite-like curing operation, with a first-order link to solid-mechanics risk through residual strain and stress indicators.

It should not be oversold as a full structural finite-element cure-stress analysis.

3 Physical background

A radiant lamp heats a material surface through electromagnetic radiation. The absorbed energy raises the temperature near the exposed surface. Heat then diffuses through the thickness by conduction, while all exposed surfaces lose heat to the surrounding air by convection and thermal radiation. For a thermoset-like material, the local temperature history drives a chemical reaction. This reaction can release heat and changes the degree of cure α from 0 to 1.

The central idea is therefore a coupled thermo-chemical problem:

$$\text{lamp flux} \longrightarrow T(x, y, z, t) \longrightarrow \alpha(x, y, z, t) \longrightarrow \text{cure quality and mechanical risk.} \tag{1}$$

This is significantly different from merely checking whether a temperature setpoint is reached. A point can become hot but remain under-cured if the temperature history is too short, and a surface can overheat while deeper layers remain colder. This is why both thermal fields and cure fields are modeled.

4 Synthetic geometry and active zone definition

The geometry is a rectangular coupon with dimensions:

$$L_x = 120\,\text{mm}, \qquad L_y = 80\,\text{mm}, \qquad L_z = 6\,\text{mm}. \tag{2}$$

The active zone is not chosen after looking at the results. It is defined before the simulation as a central ellipse representing the synthetic bonded or functional patch:

$$\left(\frac{x - x_c}{r_x}\right)^2 + \left(\frac{y - y_c}{r_y}\right)^2 \leq 1, \tag{3}$$

with

$$x_c = 60\,\text{mm}, \qquad y_c = 40\,\text{mm}, \qquad r_x = 41\,\text{mm}, \qquad r_y = 27\,\text{mm}. \tag{4}$$

This definition is geometric and independent of the computed temperature or cure fields. The global domain is still analyzed, but feasibility is judged on the active zone because it represents the functional region of interest.

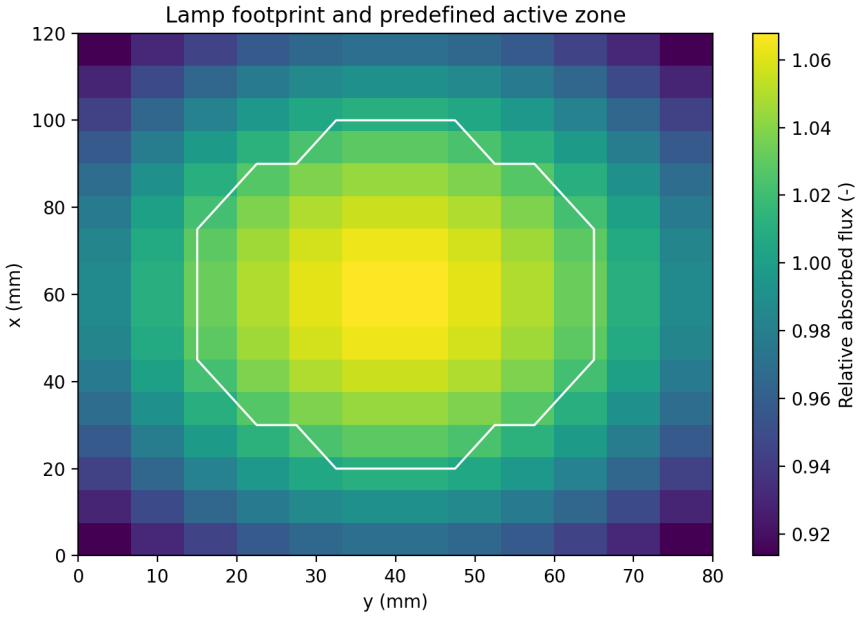


Figure 1: Synthetic lamp footprint and predefined active zone. The white contour is the geometric active zone; it is not derived from the cure result.

5 3D orthotropic heat-transfer model

The material is modeled as simplified orthotropic, more precisely transversely isotropic for heat conduction, with

$$k_x = k_y \neq k_z. \tag{5}$$

This choice is more precise than the generic word "anisotropic". A fully anisotropic model would require a complete conductivity tensor, whereas the present synthetic model uses axis-aligned conductivities.

The governing equation is

$$\rho c_p \frac{\partial T}{\partial t} = k_x \frac{\partial^2 T}{\partial x^2} + k_y \frac{\partial^2 T}{\partial y^2} + k_z \frac{\partial^2 T}{\partial z^2} + q_{rxn}, \tag{6}$$

where T is temperature in degrees Celsius for the numerical implementation, ρ is density, c_p is heat capacity and q_{rxn} is the source term from cure exotherm.

The thermal diffusivities are

$$a_x = \frac{k_x}{\rho c_p}, \qquad a_y = \frac{k_y}{\rho c_p}, \qquad a_z = \frac{k_z}{\rho c_p}. \tag{7}$$

6 Boundary conditions: radiant input and losses

The top surface receives an absorbed radiant flux

$$q_{lamp}(x, y, t) = \eta q_0(t) f(x, y), \tag{8}$$

where q_0 is the incident lamp flux, η is the absorption efficiency and $f(x, y)$ is the normalized lamp footprint.

The same top surface also loses heat through convection and thermal radiation:

$$q_{loss} = h(T_s - T_\infty) + \varepsilon\sigma \left(T_s^4 - T_\infty^4\right). \tag{9}$$

In this equation temperatures must be converted to Kelvin inside the radiation term. The net top boundary flux is then

$$q_{top} = q_{lamp} - q_{loss}. \tag{10}$$

Other exposed faces are modeled with convection and radiation losses only. This is a simplification: real tooling contact, insulation or local shielding would require additional boundary modeling.

7 Cure kinetics and exotherm

The degree of cure α satisfies

$$0 \leq \alpha \leq 1. \tag{11}$$

The model uses a Kamal-Sourour kinetic law:

$$\frac{d\alpha}{dt} = (k_1 + k_2\alpha^m)\,(1 - \alpha)^n, \tag{12}$$

with Arrhenius temperature dependence

$$k_j(T) = A_j \exp\left(-\frac{E_j}{RT}\right), \qquad j = 1, 2. \tag{13}$$

This model is widely used as a phenomenological representation of thermoset cure kinetics. In the synthetic model, the kinetic parameters are not identified from experiments; they are selected to produce realistic-looking qualitative behavior. This must be kept explicit.

The exothermic source term is approximated as

$$\left(\frac{\partial T}{\partial t}\right)_{rxn} = \frac{w_r \Delta H}{c_p} \frac{d\alpha}{dt}, \tag{14}$$

where w_r is the resin mass fraction and ΔH is the reaction enthalpy per unit resin mass.

8 Synthetic parameter table

Table 1: Main synthetic physical and process parameters.

Parameter	Symbol	Value	Role
Density	ρ	$1350\,\mathrm{kg\,m^{-3}}$	thermal inertia
Heat capacity	c_p	$1100\,\mathrm{J\,kg^{-1}\,K^{-1}}$	energy storage
In-plane conductivity	$k_x = k_y$	$0.55\,\mathrm{W\,m^{-1}\,K^{-1}}$	lateral diffusion
Through-thickness conductivity	k_z	$0.42\,\mathrm{W\,m^{-1}\,K^{-1}}$	depth diffusion
Convection coefficient	h	$12\,\mathrm{W\,m^{-2}\,K^{-1}}$	air losses
Emissivity	ε	0.82	radiative losses
Absorption efficiency	η	0.72	absorbed fraction of incident lamp flux
Ambient temperature	T_∞	22 °C	environment
Maximum temperature	T_{max}	130 °C	process constraint
Maximum centerline gradient	ΔT_{max}	20 °C	quality constraint
Mean cure target	$\bar{\alpha}_{min}$	0.90	cure constraint
Active-zone cure target	$\alpha_{active,min}$	0.86	functional-zone constraint

9 Numerical method and stability

The heat equation is discretized by finite differences on a Cartesian grid. For an interior cell, the orthotropic diffusion term is approximated by

$$\frac{dT_{i,j,k}}{dt} = a_x \frac{T_{i+1,j,k} - 2T_{i,j,k} + T_{i-1,j,k}}{\Delta x^2} \tag{15}$$

$$+ a_y \frac{T_{i,j+1,k} - 2T_{i,j,k} + T_{i,j-1,k}}{\Delta y^2} \tag{16}$$

$$+ a_z \frac{T_{i,j,k+1} - 2T_{i,j,k} + T_{i,j,k-1}}{\Delta z^2} + \left(\frac{dT}{dt}\right)_{boundary} + \left(\frac{dT}{dt}\right)_{rxn}. \tag{17}$$

An explicit time integration is used. A conservative stability condition for 3D anisotropic diffusion is

$$\Delta t \leq \frac{1}{2\left(a_x/\Delta x^2 + a_y/\Delta y^2 + a_z/\Delta z^2\right)}. \tag{18}$$

For the default medium mesh, the stability estimate is about 2.39 s, while the chosen time step is 1.8 s, which remains below the stability limit.

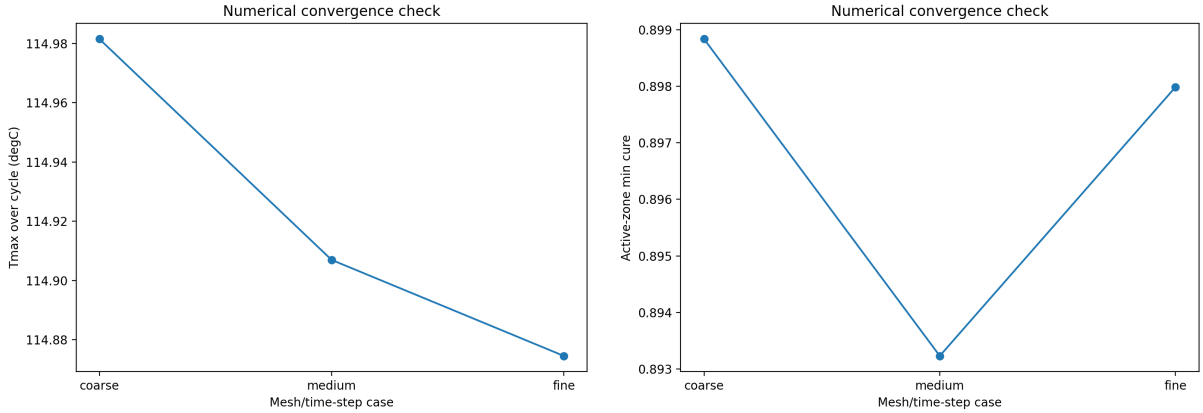
10 Convergence study

A convergence study is essential before presenting numerical values. The same optimized cycle is simulated on three mesh/time-step settings.

Table 2: Convergence study for the selected SLSQP-refined cycle. Values are rounded because the model is exploratory.

Case	Grid	Δt (s)	T_{max} (C)	$\bar{\alpha}$	$\alpha_{active,min}$	σ_{active} (MPa)
Coarse	$12 \times 8 \times 5$	2.6	115.0	0.900	0.899	2.62
Medium	$16 \times 12 \times 6$	1.8	114.9	0.902	0.893	2.69
Fine	$20 \times 14 \times 7$	1.0	114.9	0.903	0.898	2.60

The maximum temperature is very stable across the three cases. The mean degree of cure varies by less than roughly 0.4% absolute between coarse and fine. The active-zone minimum cure is somewhat mesh-sensitive but remains above the target. The global minimum cure decreases on finer meshes because corners and edge cells are resolved more sharply; this supports the decision to distinguish global and active-zone criteria.



(a) Maximum temperature.

(b) Active-zone minimum cure.

Figure 2: Convergence of two key outputs.

11 Optimization problem

The decision variables are the incident flux and lamp-on time:

$$x = (q_0, t_{on}). \tag{19}$$

The objective balances time, energy and mechanical risk:

$$J(x) = t_{on,min} + w_E E_{abs}(x) + w_\sigma \sigma_{active,max}(x), \tag{20}$$

where $t_{on,min}$ is in minutes, E_{abs} is absorbed energy per unit area and $\sigma_{active,max}$ is the active-zone stress indicator.

The constraints are

$$\bar{\alpha}(t_f) \geq 0.90, \tag{21}$$

$$\alpha_{active,min}(t_f) \geq 0.86, \tag{22}$$

$$T_{max,cycle} \leq 130\,^\circ\mathrm{C}, \tag{23}$$

$$\Delta T_{centerline,max} \leq 20\,^\circ\mathrm{C}. \tag{24}$$

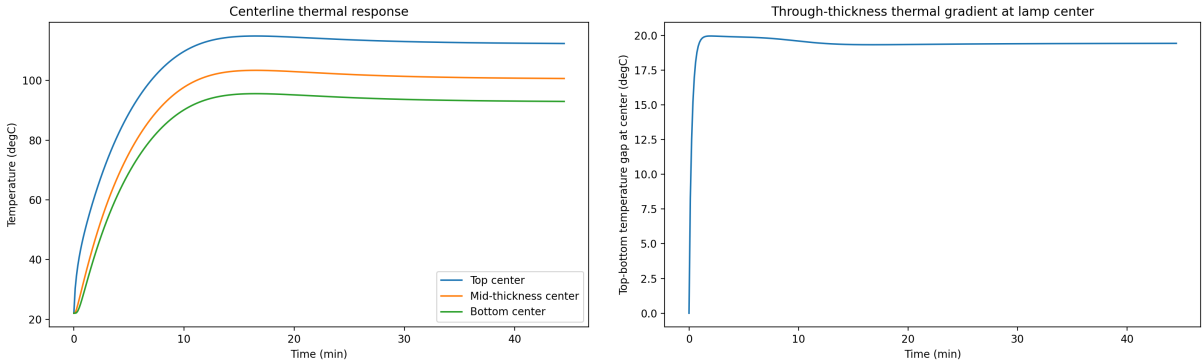
A coarse grid search is first used to identify a feasible region. Then SLSQP refines q_0 and t_{on} continuously. This is stronger than a grid-only approach, but it is still not a full optimal-control problem. A future version could optimize a piecewise power profile $q_0(t)$.

Table 3: Grid initialization versus SLSQP-refined cycle.

Method	q_0 (W/m^2)	Duration (min)	Energy (kJ/m^2)	T_{max} (C)	$\alpha_{active,min}$
Best grid	4000	52.5	9072	113.1	0.915
SLSQP refined	4095	44.5	7871	114.9	0.893

The SLSQP cycle is shorter and uses less absorbed energy while remaining close to the thermal-gradient constraint. This indicates that the initial grid solution was conservative in duration.

12 Thermal and cure results



(a) Centerline temperature.

(b) Through-thickness gradient.

Figure 3: Thermal response of the selected cycle.

The centerline top temperature rises faster than the bottom temperature, as expected for surface radiant heating. The maximum centerline gradient is close to the imposed limit, which means the optimized cycle is constrained primarily by thermal non-uniformity rather than by the absolute temperature limit.

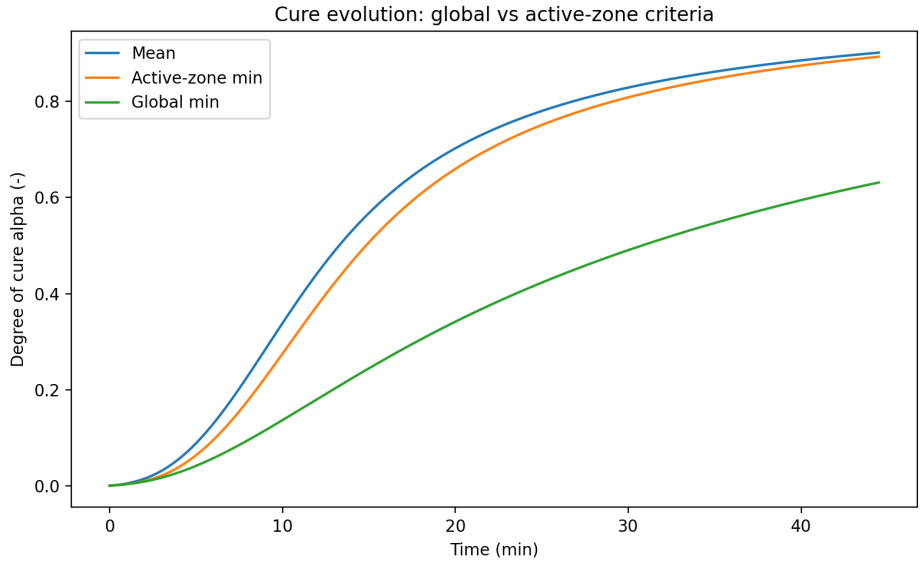
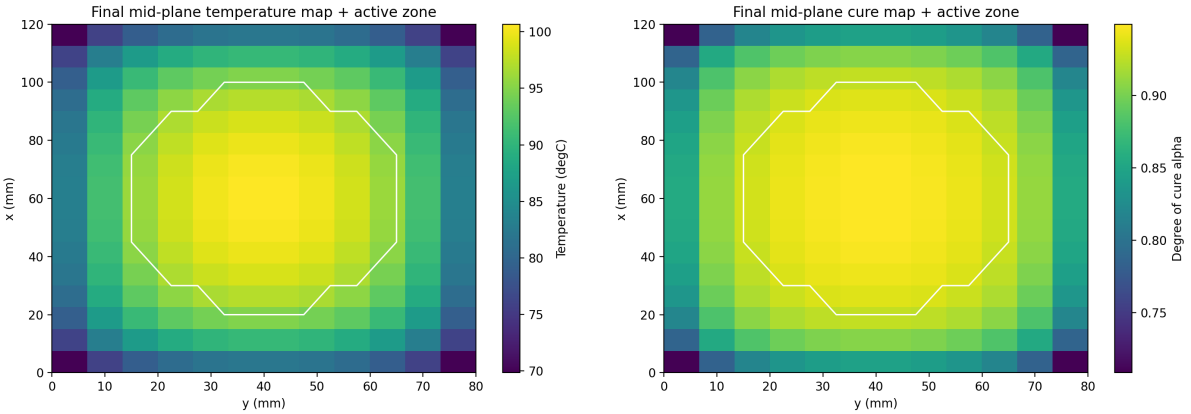


Figure 4: Cure evolution. The active-zone minimum remains much higher than the global minimum because the global domain includes cooler corners outside the functional patch.

The mean cure reaches approximately 0.902, and the active-zone minimum reaches approximately 0.893. The global minimum is much lower, around 0.63, because some outer cells receive lower flux and are not part of the functional active zone.



(a) Mid-plane temperature map.

(b) Mid-plane degree of cure map.

Figure 5: Final mid-plane fields. The active zone contour is shown in white.

13 First-order mechanical extension: residual stress indicator

V1 was purely thermo-chemical. V2 adds a first-order mechanical indicator to connect temperature/cure heterogeneity with possible residual stress risk. This does not replace a structural finite-element analysis, but it introduces the right Mechanical Engineering question: what deformation mismatch is created by the cure cycle?

The free strain is approximated as

$$\varepsilon_{free} = \varepsilon_{thermal} + \varepsilon_{cure}, \tag{25}$$

with

$$\varepsilon_{thermal} = \alpha_T (T - T_{ref}), \tag{26}$$

$$\varepsilon_{cure} = -\beta_c \max\left(0, \frac{\alpha - \alpha_{gel}}{1 - \alpha_{gel}}\right). \tag{27}$$

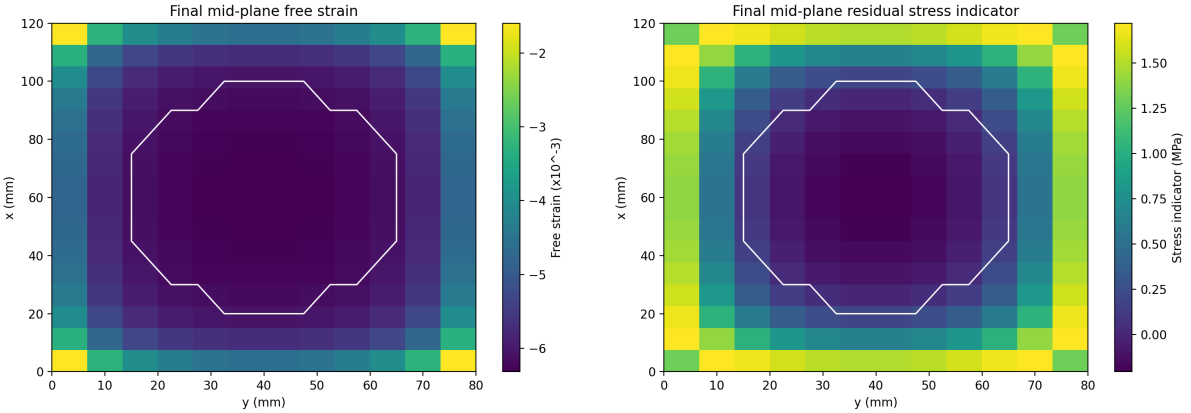
A cure-dependent modulus is introduced in a CHILE-like way:

$$E_{eff}(\alpha) = E_{uncured} + (E_{cured} - E_{uncured}) \left[\max\left(0, \frac{\alpha - \alpha_{gel}}{1 - \alpha_{gel}}\right)\right]^p. \tag{28}$$

The stress indicator is then based on mismatch with the active-zone mean free strain:

$$\sigma_{ind} = c_f \frac{E_{eff}}{1 - \nu} \left(\varepsilon_{free} - \langle \varepsilon_{free} \rangle_{active}\right). \tag{29}$$

This is not a validated residual stress prediction. It is a ranking/visualization indicator showing where non-uniform cure and temperature could create mechanically critical mismatch.



(a) Free strain field.

(b) Residual stress indicator.

Figure 6: First-order mechanical extension. The field is a stress-risk indicator, not a full structural calculation.

The maximum active-zone stress indicator is approximately 2.7 MPa in the synthetic case. This value should not be interpreted as an experimental stress value; its purpose is to extend the project from heat transfer and cure kinetics toward solid-mechanics reasoning.

14 Validation strategy

A real validation plan would require at least three levels of measurement.

Table 4: Possible validation plan.

Quantity	Possible measurement	Purpose
Temperature history	thermocouples at surface and depth; infrared camera	validate thermal field and boundary assumptions
Degree of cure	DSC; DMA; glass-transition estimation	validate cure kinetics and final conversion
Mechanical quality	lap-shear, hardness, DMA modulus, residual deformation/warpage	validate mechanical consequences of cure

The current project does not include experimental data. This is the main limitation of the portfolio. Its strength is formal modeling; its weakness is lack of real calibration and validation.

15 Limitations

The main limitations are:

- all parameters are synthetic;
- the material is simplified as orthotropic with constant thermal properties;
- the lamp model is a smooth footprint, not a calibrated optical/ray-tracing model;
- the cure model is not fitted to DSC data;
- the explicit finite-difference method is simple and not industrial-grade;

- the residual stress indicator is first-order and not a coupled thermo-chemo-mechanical finite-element analysis;
- the active zone is meaningful only if tied to a predefined functional patch.

These limitations are acceptable for a synthetic Mechanical Engineering portfolio project, provided they are stated clearly.

16 Conclusion

The V2 project is a significant improvement over the V1 proof of concept. It is no longer only a 3D thermal/cure simulation. It now includes convergence evidence, a precise active-zone definition, continuous optimization, corrected orthotropic terminology, a transparent parameter table, and a first-order mechanical stress indicator.

The project should be described as:

> A synthetic 3D thermal-process engineering model for radiant curing, coupling orthotropic heat transfer, infrared boundary conditions, Kamal-Sourour cure kinetics, exothermic reaction heat, numerical convergence, SLSQP process optimization and a first-order residual stress indicator.

This makes it a strong complement to an Industrial Engineering / Operations Research portfolio project. It does not yet replace a validated experimental or finite-element cure-stress study, but it gives a credible Mechanical Engineering foundation.

References

[1] MIT OpenCourseWare, *Introduction to Heat Transfer*, Course 2.051, Massachusetts Institute of Technology.

[2] M. R. Kamal and S. Sourour, *Kinetics and thermal characterization of thermoset cure*, Thermochimica Acta, 1976.

[3] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. Used as background for constrained optimization thinking.

[4] SciPy Documentation, *scipy.optimize.minimize* and SLSQP method.

[5] Research literature on infrared curing simulation of thermoset composites and controlled infrared curing of composite components.

[6] Arts et Metiers coursework, 2025-2026: heat transfer, thermal radiation, convection, composite materials, elementary material mechanisms and material characterization.